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Insulin resistance: the consequence of a neuroendocrine disturbance?

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Visceral obesity in man is followed by several abnormalities in endocrine secretions, including an elevated cortisol secretion as well as hyperandrogenicity in women, and a relative hypogonadism in men, suggesting a central neuroendocrine background. A pronounced insulin resistance is a main sympton of this condition and may be at least partially following these endocrine abnormalities. For example, mimicking hyperandrogenicity in women by administration of testosterone (T) in moderate doses to female rats is followed by severe muscular insulin resistance, as well as a low type I/type II muscle fiber ratio, and a low capillary density in muscle. These findings are identical to those in visceral obese women with hyperandrogenicity and insulin resistance.

Studies of details of function and morphology in the muscles in this rat model have revealed that insulin stimulation of glucose transport, glycogen synthesis and the insulin sensitive part of the glycogen synthase is diminished at submaximal insulin concentrations. However, insulin receptor number and function, as well as the total number of glucose transporter 4 appear to be normal, supported by observations that the insulin resistance is overcome by elevated insulin concentrations. These observations suggest that normal cellular mechanisms for glucose metabolism in muscle in this model are under-utilized, and that the explanation to apparent insulin resistance may be found proximal to the machinery of the muscle cell.

The histochemical changes of the muscles may provide a clue to the understanding of the mechanism. The fiber-type change might be secondary to the prevailing hyperinsulinemia because exposure to elevated insulin levels is followed by similar changes. The decreased capillary density may, however, be of importance for the apparent insulin resistance. Insulin binds to capillary endothelium, and is then apparently transferred to the site of action at the insulin receptor on the muscle cell. Extracellular insulin concentration is lower than in circulation, and is equilibrating slowly, suggesting that this process might be a rate-limiting factor for insulin action on cells. Insulin binding to capillaries is presumably porportional to capillary density. With T-treatment capillary density and blood flow in muscle are diminished in proportion to the insulin resistance. It therefore seems possible that insulin binding to capillaries, and subsequent transcapillary transport are insufficient in T-treated female rats, leading to a decreased insulin signal to the rate-limiting steps of glucose metabolism in muscle, and an apparent 'insulin resistance'. This chain of events is summarized in Figure 1. This principle might be of importance also in other conditions where a low muscular capillarization is parallel to insulin resistance.

Keywords: insulin resistance; endocrine abnormalities; hyperandrogenicity; hypogonadism; testosterone

Insulin resistance is an important metabolic perturbation which may be followed by a number of secondary metabolic derangements and diseases.[1] Its origin is not clear, however. In this review the syndrome of visceral obesity will be utilized as a 'model' to examine pathogenetic factors for insulin resistance. This syndrome is characterized by a severe insulin resistance with a multitude of other metabolic derangements as well as hypertension. In fact, this cluster of abnormalities is congruent with a metabolic syndrome, and provides a strong risk factor for the development of non-insulin dependent diabetes mellitus (NIDDM), cardiovascular disease (CVD) and stroke.[2]

Elevated free fatty acids (FFA) have been considered a strong candidate for the pathogenesis of insulin resistance in visceral obesity. FFA concentration and turn-over have been shown to be elevated in systemic circulation.[3,4] Furthermore, it seems most likely that FFA in portal circulation may also be elevated.[5] The role of FFA in insulin resistance has been reviewed repeatedly recently,[6,7] and will therefore not be dealt with further here.

Several endocrine perturbations have now been identified in visceral obesity. These include a sensitive hypothalamo-adrenal axis, as well as perturbations in sex steroid hormone concentrations in both men and women. In men testosterone levels have been observed to be low, and in women a combination of menstrual irregularities and hyperandrogenicity is frequent. Furthermore, recently low concentrations of insulin-like growth factor I have been observed.[2]

The steroid hormone abnormalities are probably followed by insulin resistance. This is most clearly the case with increased cortisol secretion, as seen for example in Cushing's syndrome. The relationship between testosterone (T) concentrations and insulin sensitivity has, however, been surprisingly little studied. By utilizing a rat model we have been able to show that castration in male rats is followed by insulin resistance in muscle, alleviated by T substitution, while excess T is again associated with insulin resistance. These results suggest that T concentrations in a 'window' of the normal range are optimal for the maintenance of insulin sensitivity. At the level of the muscle insulin stimulation of glycogen synthesis and glycogen synthase (GS) is low, but also glucose transport inhibition has been found.[2]

The significance of T deficiency for insulin resistance in visceral obese men is indicated by intervention studies. After T substitution of visceral obese men, to normalize T concentrations, an early event is a marked improvement of their insulin sensitivity. In addition, the entire metabolic syndrome is improved with decreases in plasma lipids, blood pressure, fasting blood glucose and visceral fat mass.[8] These observations strongly suggest that T deficiency plays an important role in insulin resistance of visceral obesity and the metabolic syndrome in men.

Hyperandrogenicity in women is well-known to be statistically associated with insulin resistance in such conditions as the polycystic ovary syndrome, and it has been much debated what is the primary factor in the triad obesity, hyperandrogenicity and insulin resistance in that syndrome.[9] Obesity is not necessary for the insulin resistance of the polycystic ovary syndrome, because also non-obese women with this syndrome are insulin resistant. There are, however, arguments for the hyperandrogenism being the primary factor, followed by insulin resistance, as well as the reverse sequence of events. Based on the following observations we favour the opinion that T may in fact cause insulin resistance.[10]

Female rats, rendered moderately hyperandrogenic by administration of T become severely insulin resistant after a short period of time. Insulin resistance is seen only at submaximal insulin concentrations, and is overcome at maximal concentrations, suggesting a perturbation at the level of the insulin receptor. Insulin-stimulated glucose transport is inhibited in muscle, particularly the most insulin-sensitive muscles. Glycogen synthesis and the insulin-sensitive part of the GS are severely inhibited. Furthermore, muscle histochemistry becomes deranged, with a diminished proportion of type I, slow twitch, red muscle fibers, and an increased proportion of type II, fast twitch, white fibers, particularly highly glycolytic, type IIb fibers. In addition, capillary density is markedly diminished, both in relation to area, and in proportion to muscle fiber.[11,12]

These changes occur with T administration without observable elevations of estrogen or corticosterone levels. Dihydrotestosterone seems to be inactive. Therefore the observations suggest that T is the active factor. Furthermore, FFA and glycerol are not elevated, suggesting that insulin resistance is not mediated via FFA.[11,12]

These findings have been partly reproduced in humans. Transsexual women, given T, become insulin resistant, probably most pronounced in muscle, determined by clamp measurements (unpublished observations). Details in muscle characteristics have not been possible to determine in these women.

A low concentration of sex hormone binding globulin (SHBG), probably indicating hyperandrogenicity, is a strong, independent risk factor for the development of NIDDM in women in the general population. A low SHBG was statistically correlated with fasting plasma insulin, again showing a relationship between androgens and insulin resistance.[13] Knowing the effects of T on insulin resistance in female rats, as well as in physically normal women it is of course a possibility that the hyperandrogenicity is a primary factor causing insulin reisstance, preceding NIDDM also in the general population of women.

Hyperandrogenicity is a well-known characteristic in visceral obese women, a pre-diabetic condition.[2,3,6] We have recently also found that women with established NIDDM are hyperandrogenic.[14] Free T values were closely correlated with signs of insulin resistance. Muscle is characterized by a low porportion of type I fibers, elevated type II fibers and a low capillarization in both conditions.[14,15] Furthermore, GS activity is lower than normal.[10] Both these conditions are thus identical in descriptive terms with the female rats treated with T, except the addition of obesity in the human conditions. One might therefore consider that hyperandrogenicity may indeed by a causative factor in these conditions in analogy with the situation in rats, as well as in transsexual women.

The results obtained in the rats and the lean transsexual women suggest that hyperandrogenicity may act without obesity to cause insulin resistance. However, FFA concentrations are clearly elevated in women with visceral obesity and with NIDDM, and it seems likely that the effects of T to bring about insulin resistance may also be mediated by FFA elevations. This is particularly the case since T has been shown to be a potent amplifier of lipolysis with actions at several levels from the androgen receptor to the protein kinase A and hormone sensitive lipase.[2]

The model of T-effects in female rats is useful for further studies of details in the chain of events following T administration. These include several perturbations in muscle, including insulin resistance in both glucose transport, and, particularly, glycogen synthase activation by insulin. Furthermore, histochemical changes of both fiber composition and capillarization are found.[11] These abnormalities seem to be identical to those observed in insulin resistant women in the population, as well as in the prevalent conditions of visceral obesity and NIDDM,[2,3,14,15] making it of particular importance to find out potential causative mechanisms.

Let us first consider the histochemical changes in muscle. Both the muscle fiber changes and the low capillarization correlate closely statistically with insulin resistance. It has been thought conventionally that the muscle fiber composition change is a sign of insulin resistance, because muscles in rats with a high proportion of type I fibers are insulin sensitive, in proportion to a higher concentration of insulin receptors.[16] This may, however, not be true for the following reasons. In the experiments with T-treated female rats insulin receptor number was not diminished in spite of pronounced changes in fiber composition with a low type I, and an elevated type II proportion of fibers (unpublished). Furthermore, and of more direct importance, chronic exposure of female rats to hyperinsulinemia is also followed by an identical change of fiber composition, but in these experiments an improved insulin sensitivity was found.[17] This observation strongly suggests that muscle fiber composition changes are not a primary factor in regulation of muscular insulin sensitivity, and that instead insulin resistance may in fact, via elevated insulin concentrations, change muscle fiber composition secondarily. This might occur by regulation of myosin synthesis by insulin, a possibility which is currently studied.

The capillary changes may, however, be of importance for the regulation of muscular insulin sensitivity for the following reasons. First, capillary density is closely statistically related to muscular insulin sensitivity when both have been measured concomitantly.[11,17–19] This does, however, not tell anything about causality.

Insulin binds to capillary endothelium, described for the first time already 1969.[20,21] This binding is occurring instantly with exposure of insulin, and is proportional to the capillary density,[22] as well as to capillary blood flow of muscle (unpublished). In female rats treated with T, capillary binding is decreased in proportion to the diminished density of capillaries (unpublished).

From the site of capillary binding the insulin molecules transfer through the endothelial cells to reach the insulin receptors at the surface of the muscle cells. Measurements in lymph or in the extracellular fluid of adipose tissue, show that this is occurring rather slowly, and the extracellular concentrations never reach those in circulation.[23,24] This suggests that insulin binding, and capillary transfer are rate-limiting for insulin availability in the extracellular space, and therefore for insulin action at the level of the insulin receptor on the muscle cell. This assumption is supported by the observation that insulin action is much more closely associated with extravascular than intravascular insulin concentrations.[23] Extravascular insulin concentration is likely to be dependent on the size of the pool of insulin bound to capillary endothelium, and this in turn on capillary density. These assumptions fit well with the proportionality between capillary density and blood flow on the one hand, and insulin action on the other. For example, with increased density of capillaries after physical training,[25] and after insulin exposure,[17] insulin action is facilitated. With a low density of capillaries, insulin sensitivity is diminished in close parallelity, observed in such conditions as physical inactivity,[25] exposure of steroid hormones[2,11] as well as clinical conditions such as visceral obesity[15,19] and NIDDM.[14] It is thus possible that capillary density is rate-limiting for muscular insulin action by modifying the pool-size of the immediately available insulin for transcapillary transfer to its site of action on the muscle cell.

The model of androgenized female rats may provide further evidence for the role of transcapillary transport of insulin for the resulting insulin resistance by measurement of other potentially rate-limiting steps for insulin action. We know from experiments that both glucose transport, and, particularly GS are less active than normal in such rats. The number of insulin receptors and their function in terms of thyrosine kinase activity are, however, normal (unpublished). Furthermore, the number of glucose transporter 4 (glut 4) is also normal (unpublished). These observations, taken together, suggest that the muscular system for insulin binding, signalling to glucose transport, as well as the capacity for glucose transport and, subsequent metabolic steps are in fact normal, but under-utilized. This in turn suggests that the signal from the insulin receptor to the glucose transporters and to the GS is diminished, and that this might in fact be the primary factor. The dose-response of insulin in T-treated female rats strongly supports this assumption, because insulin resistance is only found at submaximal levels of insulin, and is overcome at elevated concentrations of insulin.[12]

These observations taken together, support the following attempt to understand the mechanisms involved, depicted schematically in Figure 1. In T-treated female rats capillarization and blood flow in muscle are diminished. This is followed by a lower than normal transfer of insulin from the capillary binding site to the extravascular space, and therefore a diminished traffic of insulin over the insulin receptor on the muscle cell surface. This in turn produces an inadequate signal to the translocation process of glut 4, as well as to the insulin sensitive part of the GS system. The results would be expected to be a diminished glucose transport, as well as a severe inhibition of glycogen synthesis. This fits with observations on the integrated parts of the chain of events of insulin action, and will result in a condition interpreted as muscular insulin resistance.

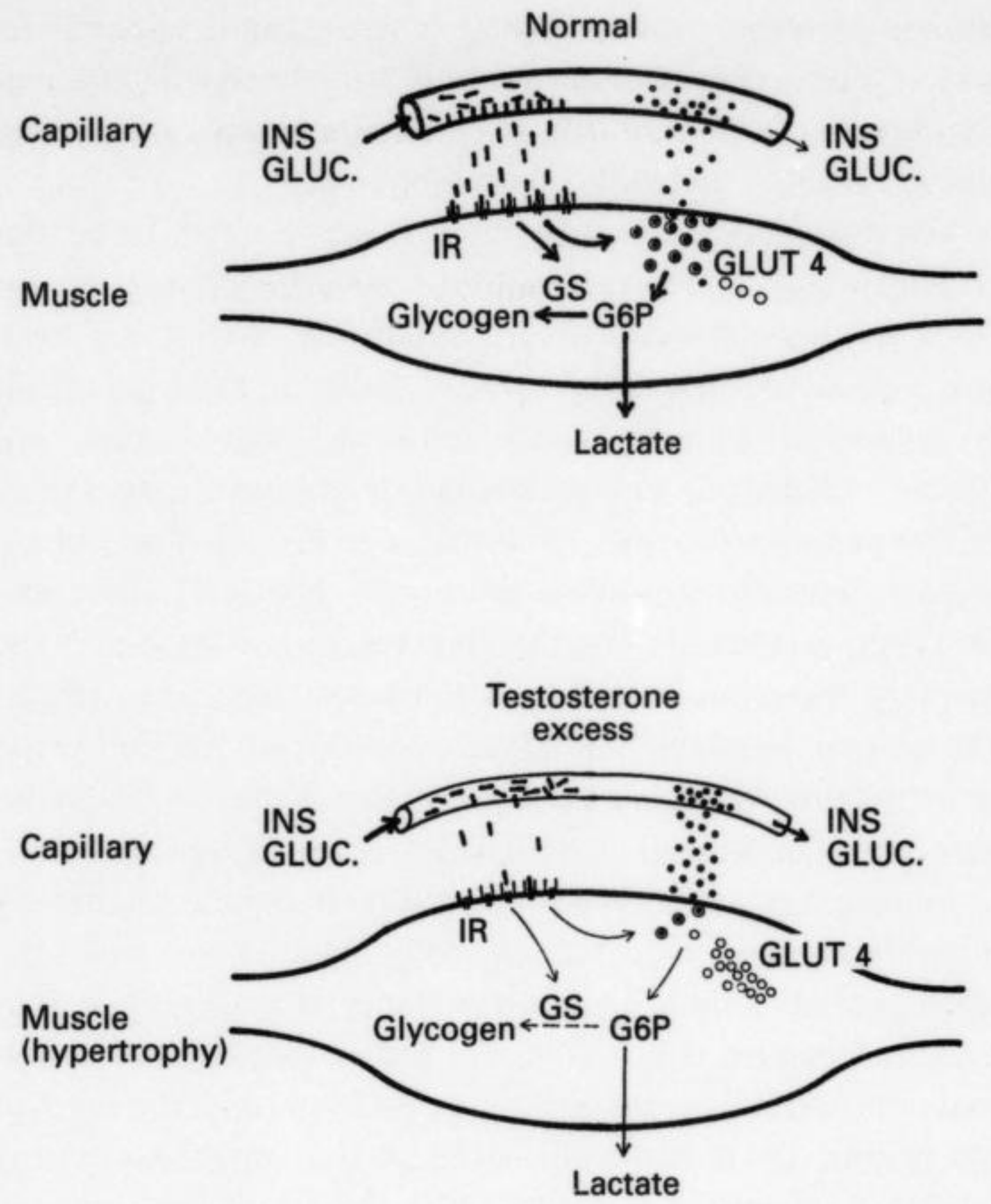

Figure 1 Schematic representation of a proposed mechanism for insulin resistance in muscle after testosterone treatment of female rats. With testosterone excess capillary density is diminished and the capacity of binding and transcapillary transport of insulin insufficient. Insulin receptor (IR) number and function are intact and the number of glucose transporter 4 (Glut 4) normal. By the decreased availability of insulin at the level of the muscle cell, the insulin signal to glucose transport and the insulin sensitive part of the glycogen synthase system (GS), glucose transport and glycogen synthesis become diminished. For further details and for references, see text. G6P: Glucose 6 phosphate.

These results seem to open up the intriguing possibility that capillary density may in fact be rate-limiting for insulin action in the model of T-treated female rats. It is of course also possible that the low capillary density in muscle in other conditions may in fact be of importance for the overall regulation of insulin sensitivity. These conditions include visceral obesity and NIDDM in women, which are hyperandrogenic conditions with a low capillary density.

The regulation of capillary density then becomes a question of considerable potential importance. Insulin itself is known to facilitate the formation of new capillary endothelial cells *in vitro*,[26] and insulinization of rats is followed by an increased capillary density.[17] Insulin-like growth factor I also influences on capillary growth, suggesting that growth hormone might be involved.[27] T-treatment is associated with a low capillary density,[11] and T may therefore also be of importance for regulation of capillarization in muscle. Studies on the potential influence of other hormones seem to be of considerable interest.

In summary, there are multiple endocrine abnormalities in visceral obesity, of which at least a relative hypercorticolism, and abnormal sex steroid hormone secretions may contribute to the creation of insulin resistance in muscle. Using a model, where administration of T to female rats is followed by severe insulin resistance, has provided some insight into the mechanisms in muscle tissue. At the level of the muscle cell the following observations have been made in these rats. The density and function of the insulin receptor appear to be normal, and the total number of glut 4 is unaffected. Nevertheless, glucose transport, and glycogen synthesis, as well as the insulin sensitive part of the GS show insufficient activities. These observations suggest, but do not show conclusively, that the cellular events in muscle are unaffected by T-treatment in this model, and that instead the insulin signal via the insulin receptor is insufficient. The perturbation leading to under-utilization of the glucose metabolism machinery in muscle may therefore be found proximal of the muscle cell.

The morphology of muscle is changed in these T-treated female rats with a lower than normal type I/type II fiber ratio, as well as a low degree of capillarization. The fiber-type changes may well be secondary to the prevailing hyperinsulinemia. The low capillary density might, however, play a role by providing an insufficient intermediary pool of insulin for subsequent transcapillary transport to its active site at the muscle cell. This possibility is supported by results of experiments where extracellular concentrations of insulin have been shown to be considerably lower than those in circulation, with a slow equilibration. Capillary insulin binding is proportional to capillary density, and diminished in proportion to the low capillary density after T-treatment of female rats. The 'insulin resistant' muscle metabolism in this model is normalized by providing the muscle with elevated concentrations of circulating insulin, suggesting that transcapillary transport of insulin is normalized by exposure of the transport system to high insulin concentrations.

In brief, transcapillary transport of insulin may be the rate-limiting factor for insulin action in this model with a low density of muscle capillaries. This mechanism might contribute to muscular insulin resistance also in other conditions with a low capillary density of muscles. This hypothesis is summarized in Figure 1.

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